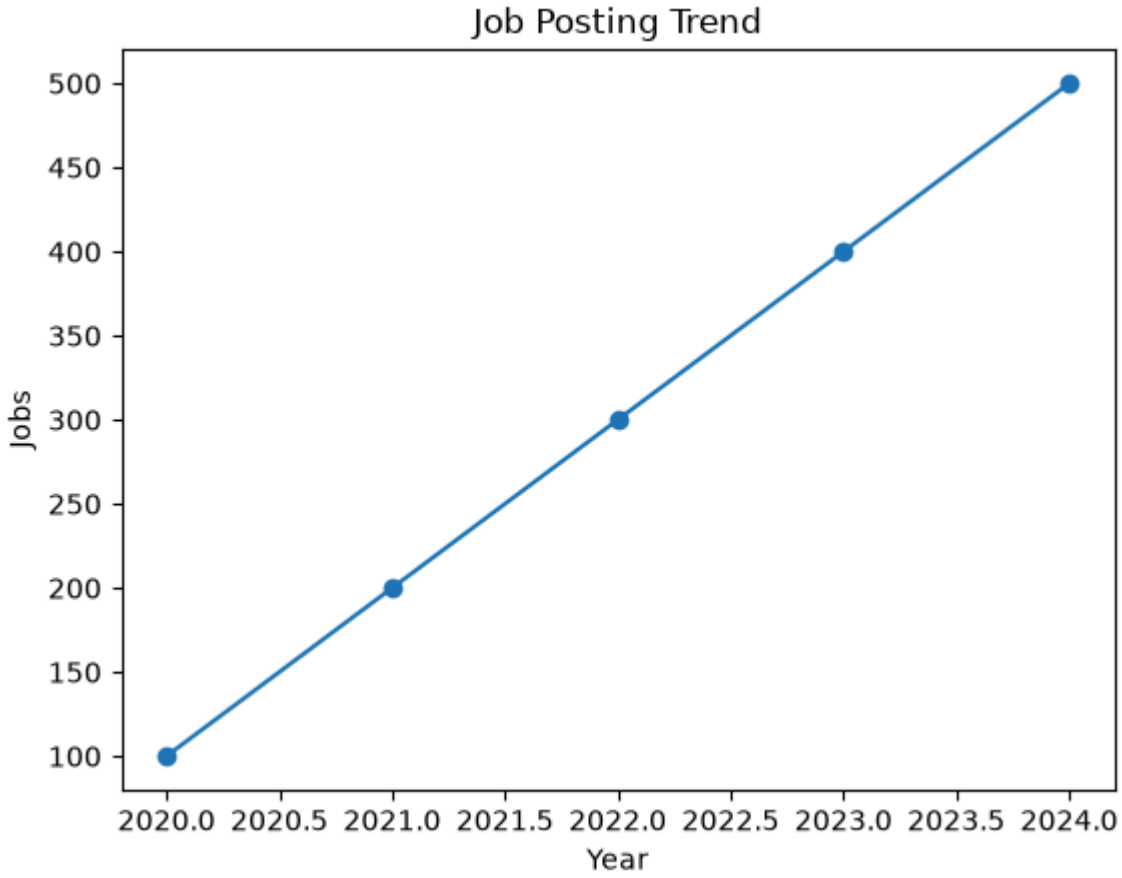
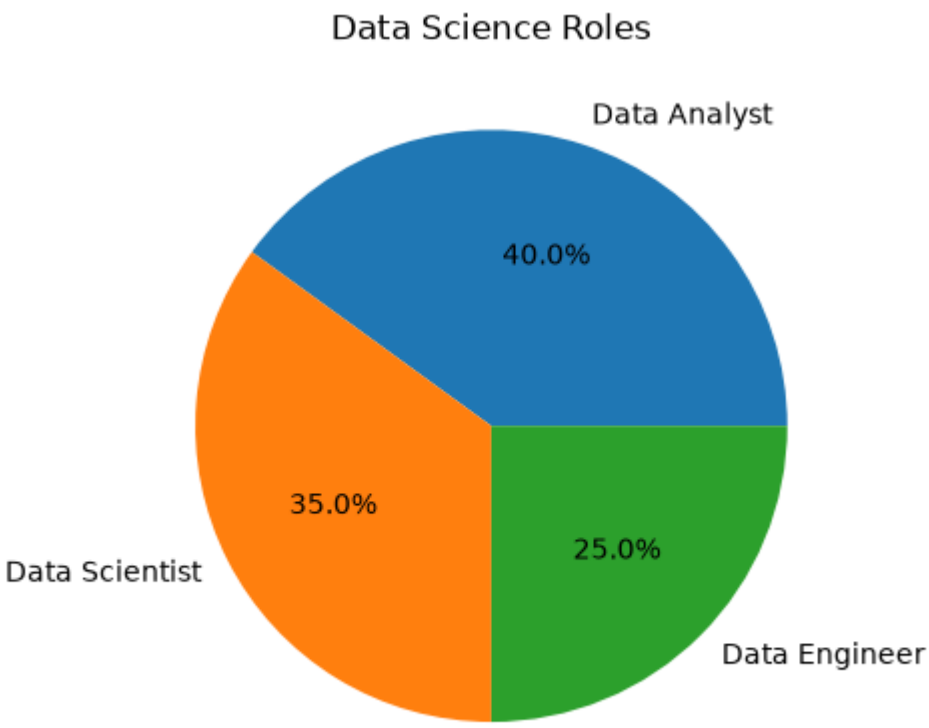
In [8]:
```
import pandas as pd
import matplotlib.pyplot as plt
df=pd.DataFrame({
    "Year" :
    [2020,2021,2022,2023,2024],
    "Jobs":
    [100,200,300,400,500]})
plt.plot(df["Year"], df["Jobs"],marker="o")
plt.title("Job Posting Trend")
plt.xlabel("Year")
plt.ylabel("Jobs")
plt.show()
```



In [14]:
```
import pandas as pd
import matplotlib.pyplot as plt
df=pd.DataFrame({
    "Role":["Data Analyst","Data Scientist","Data Engineer"],
    "Count":[40,35,25]})
plt.pie(df["Count"],
labels=df["Role"], autopct="%1.1f%%")
plt.title("Data Science Roles")
plt.show()
```



In [16]:
```
import pandas as pd
df=pd.DataFrame({"Name":
["Asha","Rahul"],"Marks":[90,85]})
print("Structured Data")
print(df)
semi={"Name":"Asha","Marks":90}
print("\nSemi-Structured Data")
print(semi)
text="Data Science is interesting."
print("\nUnstructured Data")
print(text)
```

```
Structured Data
    Name  Marks
0   Asha     90
1  Rahul     85

Semi-Structured Data
{'Name': 'Asha', 'Marks': 90}

Unstructured Data
Data Science is interesting.
```

In [18]:
```
from cryptography.fernet import Fernet
key=Fernet.generate_key()
f=Fernet(key)
msg="Hello"
enc=f.encrypt(msg.encode())
print(enc)
dec=f.decrypt(enc)
print(dec.decode())
```

```
b'gAAAAABqciD-yh-glWqBeudLZOdHY39gfsUKfbllteNKz4bMznFybQHgdcbHH8MnrQSsS9hKhQ2X5JZ
pQpqc5gkXF8BHZTG-tQ=='
Hello
```

In [ ]: